

4/22/2024

Informatics 3 Year Project

Deliverable 2: Specification

CYBER CREW



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1. Updated Problem Statement, Proposed Solution & Use Cases

1.1 Problem Statement

The current AutoWorx Barnyard's customer communication and inquiries, invoice & quotation generation, task coordination, and process documentation rely on outdated methods which include the use of telephone & WhatsApp for communication, Microsoft Word Invoice & Quotation Template, task assignment by mouth resulting in a lack of systematic customer relationship management, task delegation, and process documentation hindering productivity, customer engagement, and knowledge sharing.

1.2 Proposed Solution

We will develop a workshop management system that prioritizes customer engagement, seamless communication, and efficient service estimation costs. This system aims to improve operations by enhancing transparency, delegating mechanics' tasks, automating the quoting and invoice generation process, maintaining comprehensive client service history, and promoting skill sharing among team members to ensure consistent service quality.

1.3 Use Cases

1.3.1 User Management

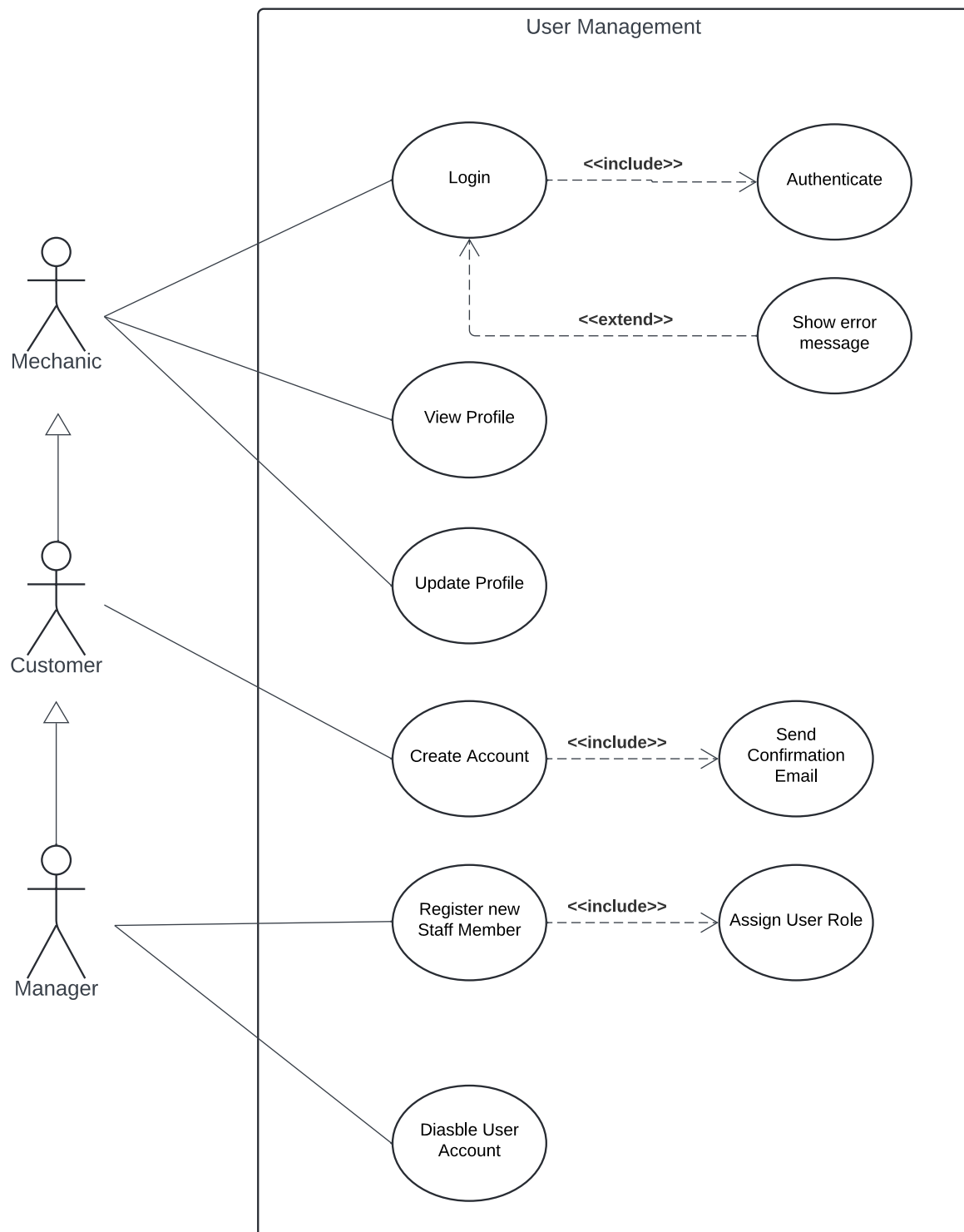


Figure 1: User Management Use Case Diagram

1.3.2 Service Request Management

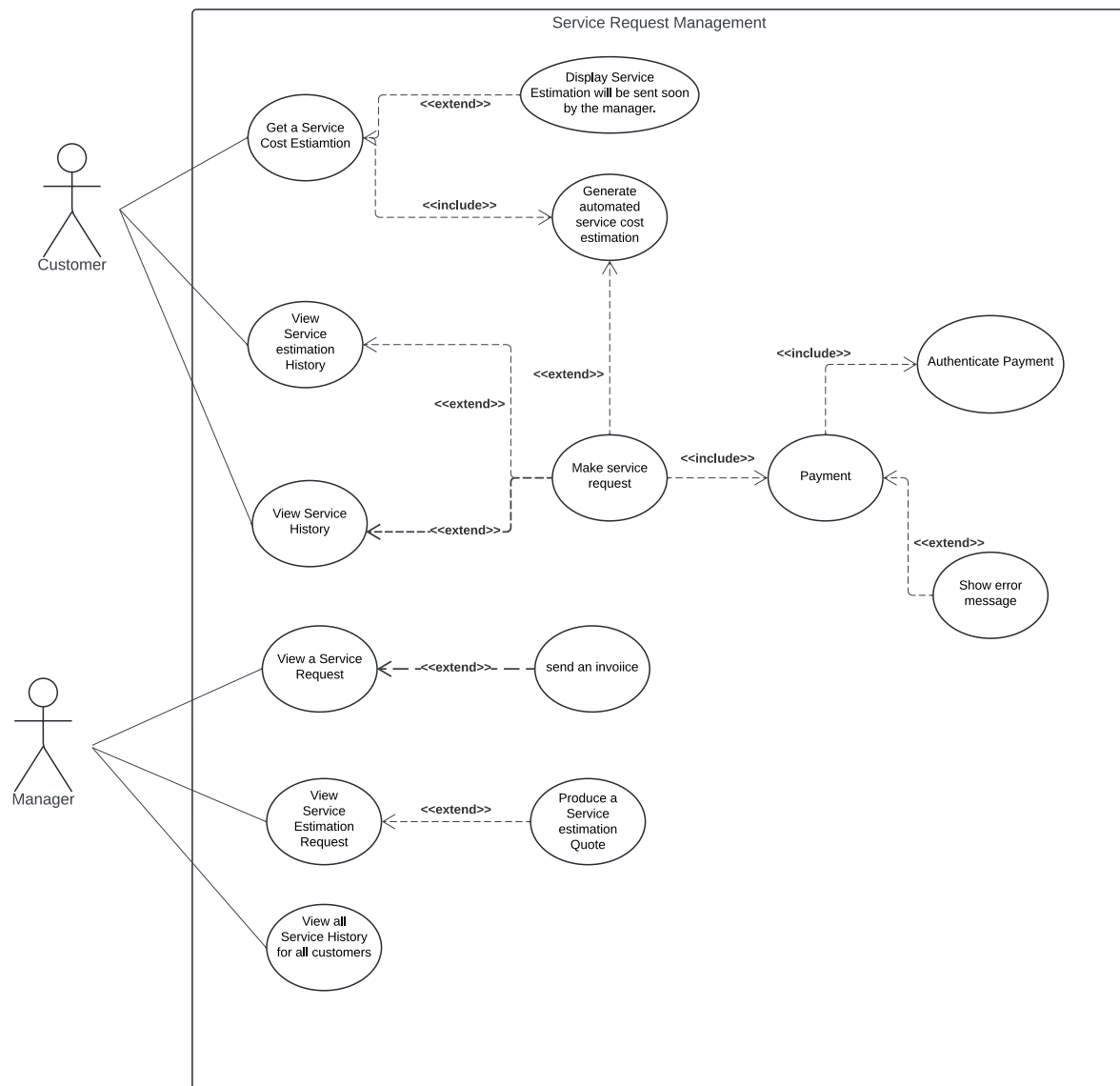


Figure 2: Service Request Management Use Case Diagram

1.3.3 Task Management

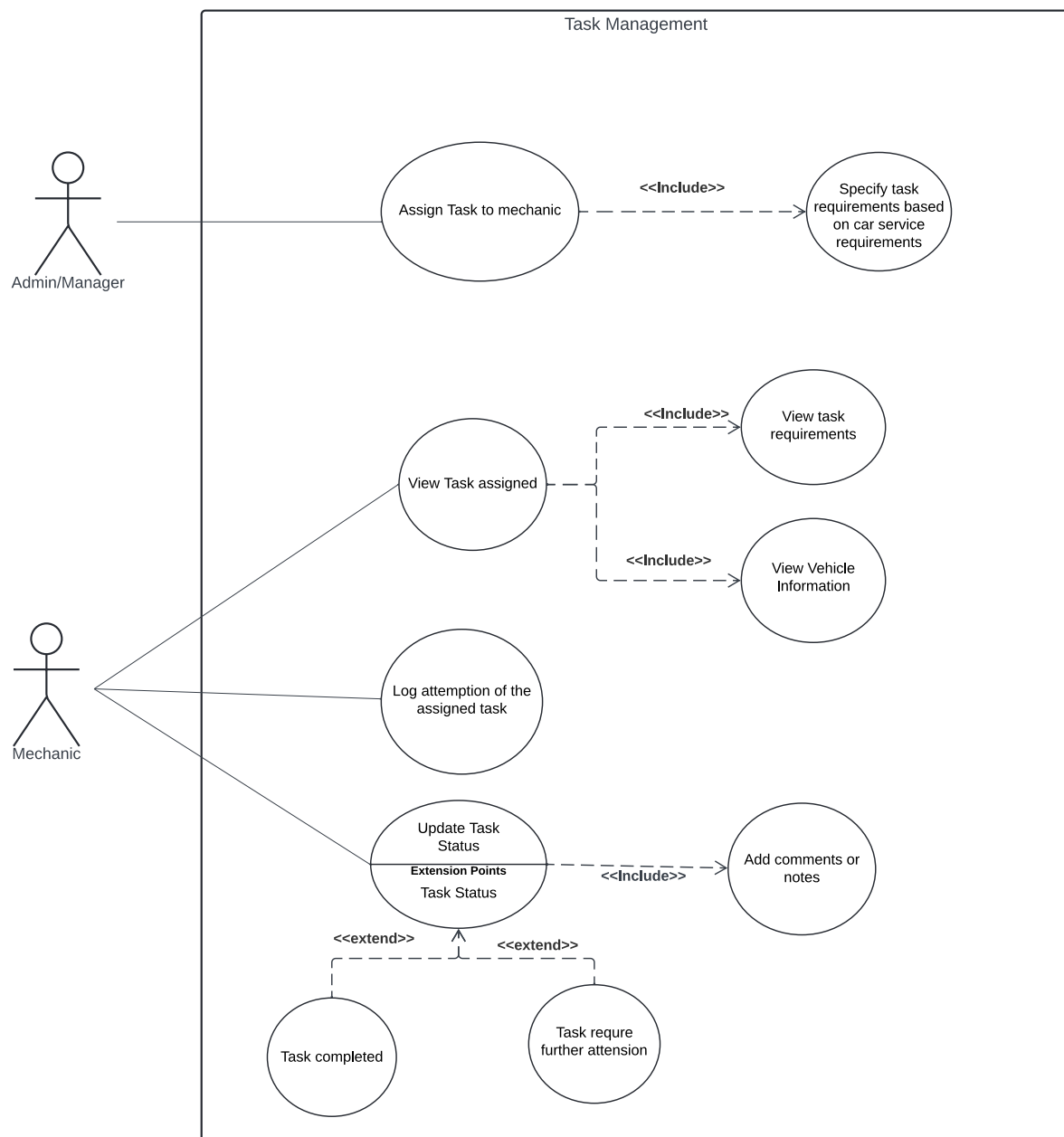


Figure 3: Figure 3: Task Management Use Case Diagram

1.3.4 Performance Monitoring

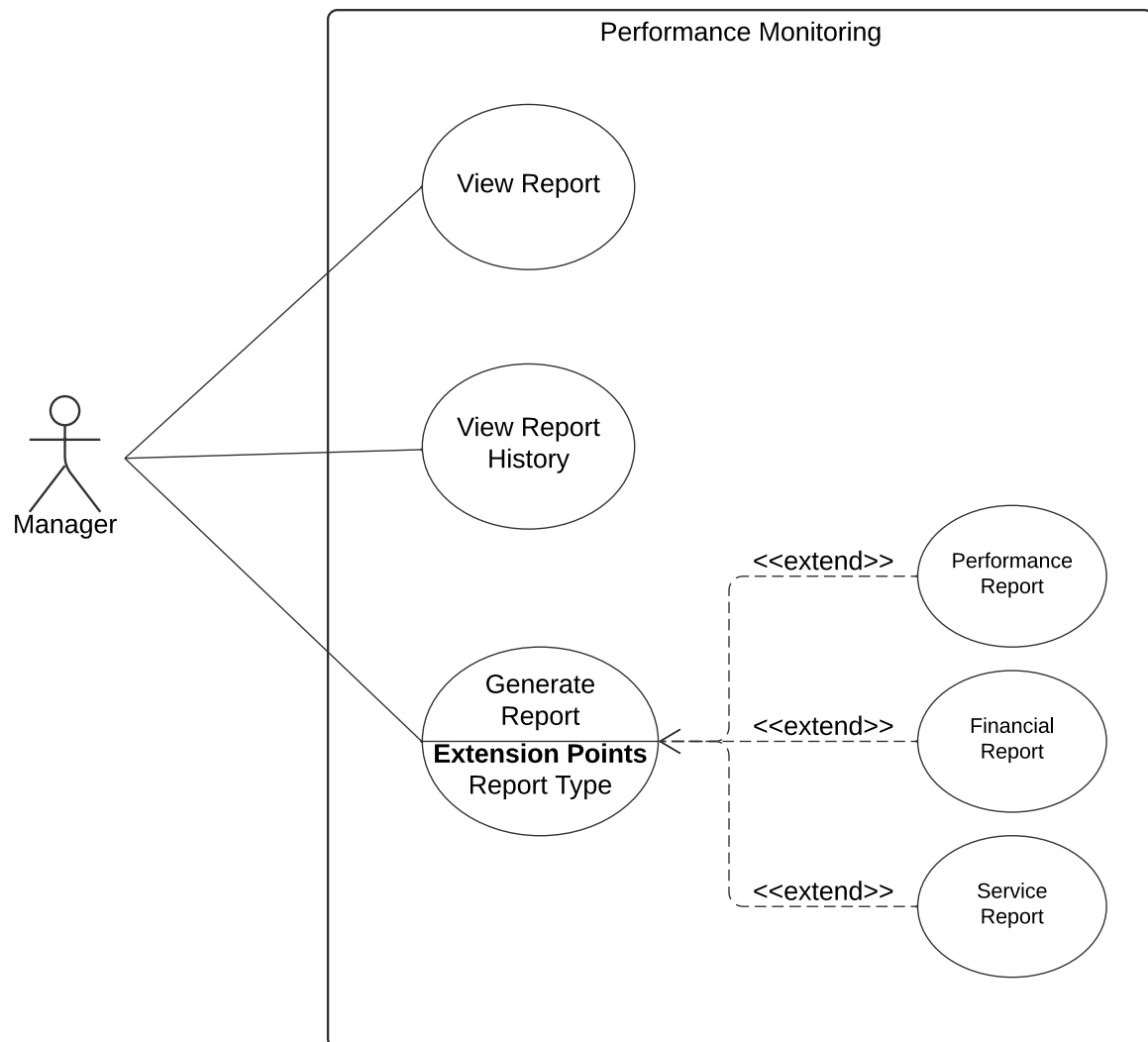


Figure 4: Skills Sharing Use Case Diagram

1.3.5 Appointment Management

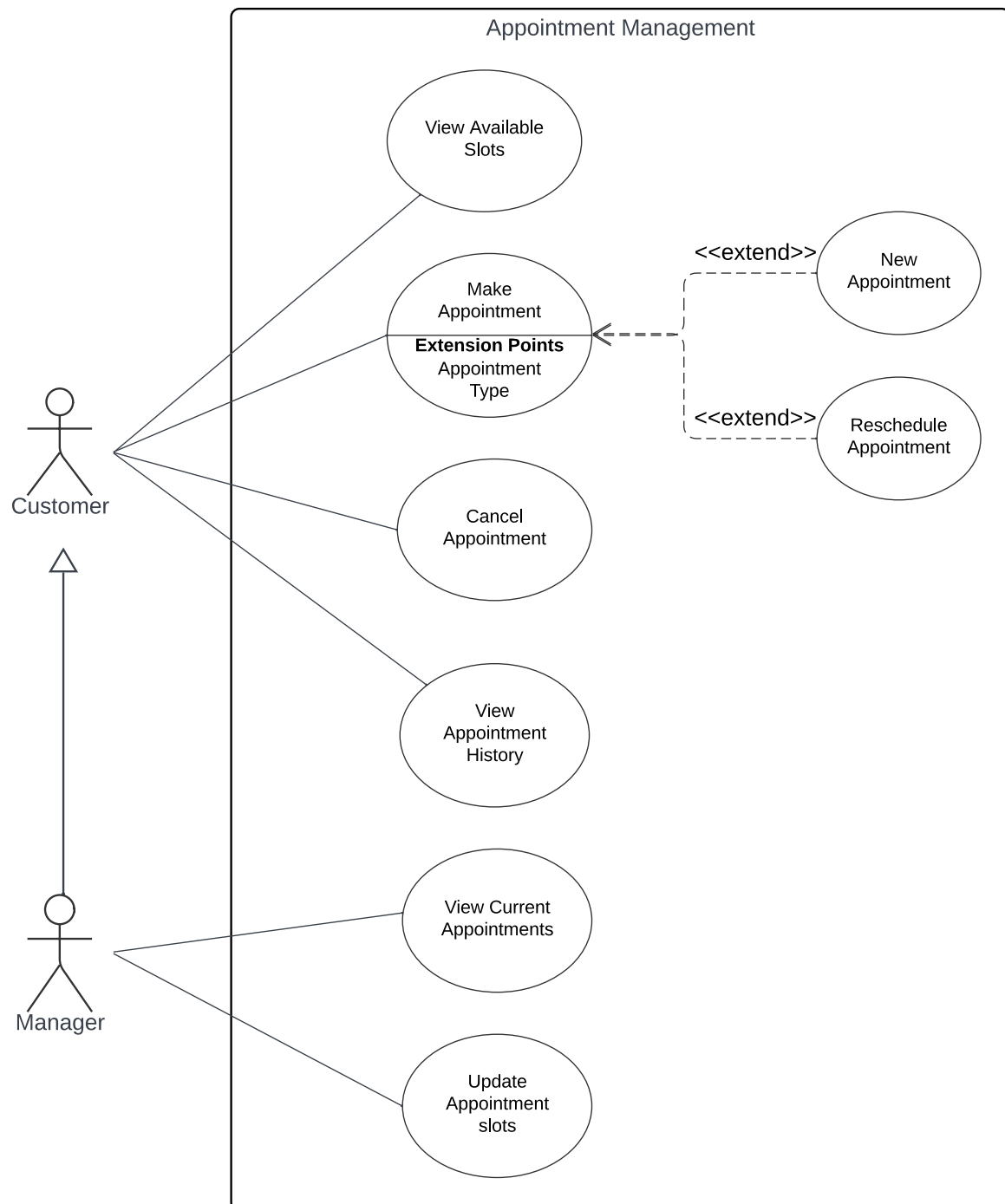


Figure 5: Appointment Management Use Case Diagram

1.3.6 Skills Sharing

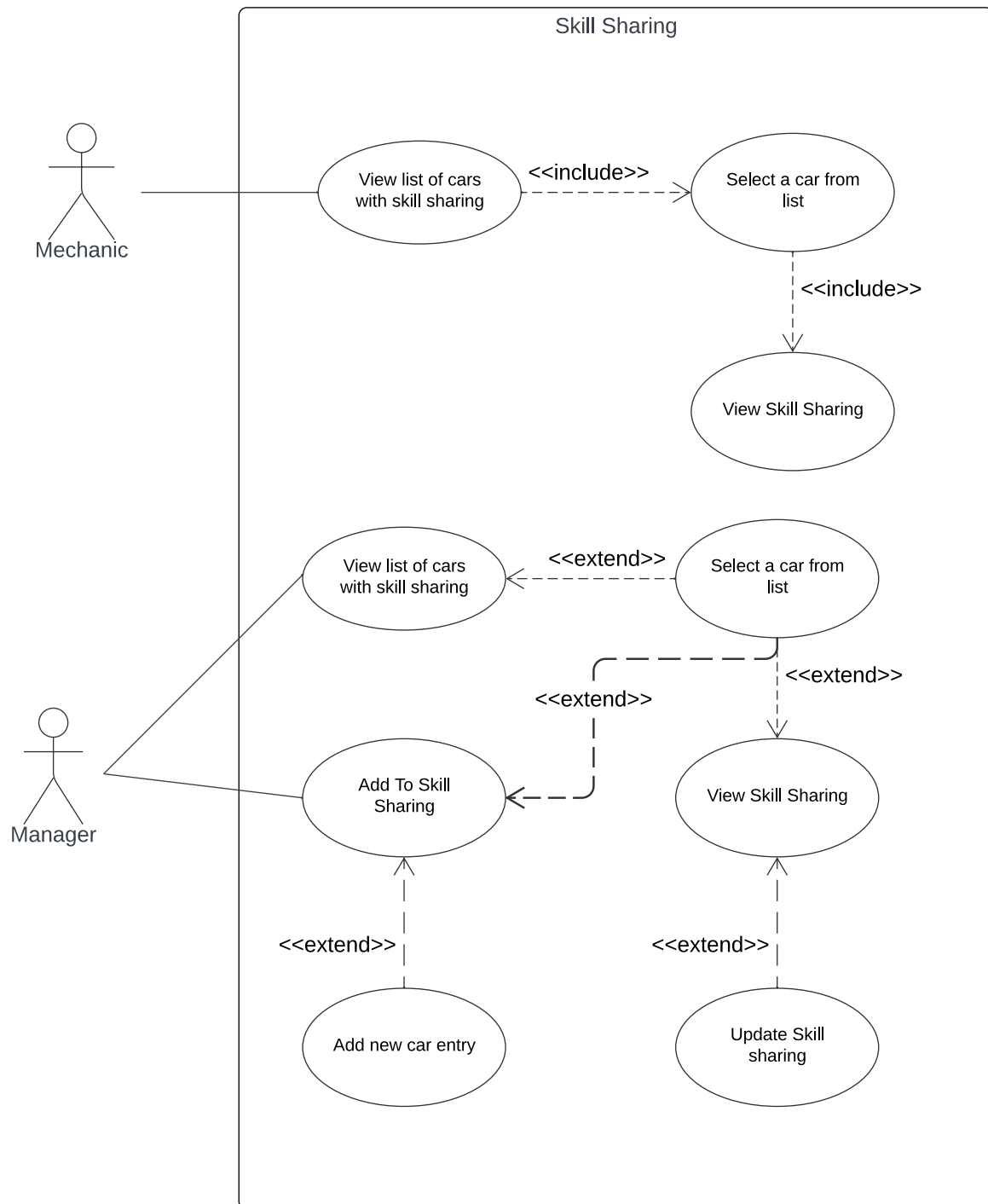


Figure 6: Skills Sharing Use Case Diagram

2. Use Case Descriptions

2.1 User Management

a) Customer Registration:

Brief Description: This use case describes the process of a customer registering for a new account on the system.

Actors: Customer

Basic Flow:

1. The customer navigates to the registration page.
2. The customer submits the registration form.
3. The system validates the entered information.
4. The system creates a new user account with the provided details and a confirmation message is sent to the user.

Alternate Flows:

3a. If the entered email already exists in the system display an error message indicating that the email is already in use.

3b. Prompt the customer to enter a different email address.

Inputs:

- Customers information.

Outputs:

- Error message: "This email is already in use. Please use a different email address."
- Confirmation message: "Your account has been created successfully."

Postconditions:

- The customer is granted access to their dashboard.

b) Registering a Mechanic

Brief Description: This use case describes the process of a manager registering a mechanic through the manager's dashboard.

Actors: Manager

Preconditions:

- The manager is logged in the system.

Basic Flow:

1. The manager accesses the dashboard.
2. The manager selects the option to register a mechanic.
3. The dashboard presents a form for the manager to enter the mechanic's details.
4. The manager fills in the required details for the mechanic.
5. The manager submits the registration form.
6. The system validates the entered information.
7. The system creates a new mechanic account with the provided details.
8. An email with login instructions, including the temporary password, is sent to the mechanic.

Alternate Flows:

6a. If the manager attempts to register a mechanic with an existing email display an error message indicating that the email is already in use.

6b. Prompt the manager to enter a different email address for the mechanic.

Inputs:

- Mechanic's information.
- Temporary password for the mechanic.

Outputs:

- Error message: "This email is already in use. Please use a different email address."
- Confirmation message: "Mechanic registered successfully. An email with login instructions has been sent to the mechanic."

Postconditions:

- A new mechanic account is created in the system with the provided details.
- An email with login instructions, including the temporary password, is sent to the mechanic.

c) User Login:

Brief Description: This use case describes the process of a user logging into their account.

Actors: Customer, Manager, Mechanic

Preconditions:

- The user is registered on the system.

Basic Flow:

1. The user navigates to the login page.
2. The user submits their login credentials.
3. The system validates the credentials against stored user data.
4. The system grants access to the user's dashboard.

Alternate Flows:

3a. If the credentials are incorrect display an error message indicating that the credentials are incorrect.

3b. Prompt the user to re-enter their credentials.

Inputs:

- User's login credentials.

Outputs:

- Error message: "Incorrect email or password. Please try again."
- Access granted to user dashboard.

Postconditions:

- The user is granted access to their dashboard.

d) Managing Profile:

Brief Description: This use case describes the process of a user managing their profile information.

Actors: Customer, Manager, Mechanic

Basic Flow:

1. The user accesses the dashboard by logging in.
2. The dashboard displays the user's current profile information.
3. The user selects the option to edit their profile.
4. The dashboard presents a form with editable fields for the user to update their profile information.
5. The user makes changes to their profile information.
6. The user submits the changes.
7. The system validates the changes.
8. The system updates the user's profile accordingly.
9. The dashboard displays a confirmation message indicating successful profile update.

Alternate Flow:

7a. If the user enters invalid changes prompt the user to correct the errors and resubmit the changes.

Inputs:

- User's credentials
- Updated profile information

Outputs:

- Confirmation message: "Your profile has been updated successfully."
- Error message: "Invalid changes. Please correct changes."

Postconditions:

- The user's profile information is updated in the system according to the changes submitted.

2.2 Service Request Management

a) Service Estimation Request:

Brief Description: This use case describes how a customer requests for a service estimation that is auto generated.

Actor: Customer

Preconditions: Customer is on the quote request page (does not have to be logged in).

Basic Flow:

1. Customer fills a form specifying services that they require.
2. Customer sends the request for a quote.
3. Customer receives an auto generated service estimation quote.
4. Customer can request for a service if they are satisfied with the costs.

Alternate Flow:

1a. If the service required by the customer is not listed, they should be able to specify the service they require.

3a. If the quote is not generated, customer will receive a message stating that the request for a service request has been sent to the manager and they will be notified as soon as the manager sends the service estimation quote.

Inputs:

- Customer's service requirements filled in the form.
- Customer's request for a quote.

Outputs:

- Auto-generated service estimation quote.
- Notification of successful quote generation or a message stating that the request has been sent to the manager for manual estimation.

Postcondition:

- The customer receives either an auto-generated service estimation quote or a notification that their request has been forwarded to the manager.
- If an auto-generated quote is provided, the customer can proceed to request the service if satisfied with the costs.

b) Manage service estimation quote request:

Brief Description: This use case describes how a manager manages a service estimation quote request from a manager.

Actor: Manager

Preconditions: Manager is logged in to the admin website and is on the manage quotes page.

Basic Flow:

1. Manager goes through list of service estimation quote requests.
2. Manager selects one of the service estimation quote requests.
3. Manager generates quotation for the client with available date.

Inputs:

- List of service estimation quote requests.
- Manager's selection of a specific quote request.

Outputs:

- Generated quotation for the selected quote request with available dates.

Postcondition:

- The manager successfully generates a quotation for the selected service estimation quote request.

c) View quote:

Brief Description: This use case describes how a customer can view a list of service request they have done in the past and use it to request for a service.

Actor: Customer

Preconditions: Customer is logged on the app and is on the manage quotes page.

Basic Flow:

1. Customer views generated quote with associated costs.
2. Customer can request for a service if they are satisfied with the costs.

Inputs:

- List of generated quotes with associated costs.

Outputs:

- Ability for the customer to view the generated quote.
- Option for the customer to request a service if satisfied with the costs.

Postcondition:

- The customer views the generated quote and can proceed to request a service if satisfied.

d) Sending an invoice:

Brief Description: This use case describes how a manager can send an invoice to the customer.

Actor: Manager and Customer

Preconditions: Manager is logged in to the admin website and is on the Service request page.

Basic Flow:

Actor: Manager

1. Manager goes through list service requests.
2. Manager selects one of the service requests.
3. Manager sends an invoice to the customer after the service request is made.

Alternate Flow:

3a. If service request from client is missing any information, then manager sends a reply to the client specifying what is missing.

Inputs:

- List of service requests.
- Selection of a specific service request.
- Necessary information to generate the invoice (e.g., service details, customer information).

Outputs:

- Sending of an invoice to the customer.
- If necessary, a reply specifying missing information if the service request is incomplete.

Postcondition:

- The manager successfully sends an invoice to the customer after the service request is made.

e) Service History:

Brief Description: This use case describes how a manager/Customer can access car service history.

Actors: Manager and Customer

Preconditions: Manager is logged in to the admin website and is on the manage quotes page.

Basic Flow:

1. Customers can view service history of their vehicles.
2. Customer can request for the same service again.
3. Managers can view service history of all vehicles.

Inputs:

- Access to service history data for vehicles.
- Request from the customer or manager to view service history.
- Identification of the vehicle (for customer) or all vehicles (for manager).

Outputs:

- Display of service history for the specified vehicle(s).
- Ability for the customer to request the same service again if desired.

Postcondition:

- Customers and managers can view the service history of vehicles, facilitating future service requests based on past experiences.

2.3 Task Management:

a) View Task:

Brief Description: This use case describes the process of Mechanics viewing the tasks assigned to them.

Actor: Mechanic

Preconditions: Mechanic must be logged into the system.

Basic Flow:

1. Mechanics log into their dashboard of the mobile app.
2. Mechanics navigate to the "Tasks" section.
3. Mechanics view the list of assigned tasks.
4. Each task displays service requirements and vehicle information.
5. Mechanics can click on a task to view additional details if needed.

Alternate Flows:

5a. If there are no assigned tasks, the system displays a message indicating that there are no tasks currently assigned to the mechanic.

Inputs:

- Mechanic's login credentials.
- Click a task on the "Tasks" section.

Outputs:

- List of assigned tasks with relevant details (service requirements, vehicle information).
- Additional details of the selected task (if clicked).

Postconditions: Mechanics have successfully viewed assigned tasks with relevant details.

b) Update Task:

Brief Description: This use case describes the process of Mechanics updating the status of assigned tasks to reflect their progress or completion.

Actor: Mechanic

Preconditions: Mechanics must be logged into the system and have assigned tasks.

Basic Flow:

1. Mechanics log into their mobile app dashboard.
2. Navigate to the "Tasks" section.
3. Select the task they want to update.
4. Update the task status by marking it as in progress, completed, or requiring further attention.
5. Mechanics have the option to add comments or notes to the task update.
6. Mechanics save/submit the changes.

Alternate Flows:

3a. If there are no assigned tasks or the selected task is not found, the system displays a "No available tasks at the moment/ No tasks found" message.

4a. If the mechanic tries to update a task without selecting one, the system prompts them to choose a task first.

Inputs:

- Mechanic's login credentials.
- Click a task tab to navigate to "Tasks" section.
- Selection of the task to update.
- Updated status of task (in progress, completed, requiring further attention).
- Optional comments or notes.

Outputs:

- Updated task status and any added comments/notes.
- Confirmation message upon successful submission.

Postconditions: Task status and any comments/notes are successfully updated and reflected in the system in real-time.

c) Assign Task:

Brief Description: This use case describes the process of Managers assigning tasks to mechanics, providing necessary details and deadlines of that particular task.

Actor: Managers

Preconditions: Managers must be logged into the system.

Basic Flow:

1. Managers log into their web application dashboard.
2. Managers navigate to the "Tasks" section.
3. Selects a task that needs to be done based on the service requested by customer.
4. Select a mechanic to assign the task to.
5. Specify additional task details such as service requirements, vehicle information, and deadlines.
6. Managers confirm and submits the task assignment.

Alternate Flows:

4a. If no mechanics are available for assignment, the manager receives a notification and is prompted to assign tasks later.

Inputs:

- Manager's login credentials.
- Click a task on the "Tasks" section.
- Selection of the task to be assigned.
- Selection of the mechanic to assign the task to.
- Additional task details (service requirements, vehicle information, deadlines).

Outputs:

- Confirmation message upon successful assignment of the task.

Postconditions: The task is successfully assigned to the selected mechanic with specified details and deadlines.

2.4 Performance Monitoring Management:

a) Generate Report:

Brief Description: This use case describes the process of Managers generating reports on key metrics to assess the performance of the workshop.

Actor: Manager

Preconditions: Manager must be logged into the system.

Basic Flow:

1. Manager navigates to the "Reports" section.
2. Selects the type of report to generate (e.g., task completion rates, service turnaround times, customer feedback scores/ratings).
3. Specifies the time period/frame for the report.
4. System processes the request and generates the report based on the selected metrics and time period.
5. Manager receives the generated report.

Alternate Flows:

3a. If there is no data available for the specified time period or metric, the system notifies the manager and suggests adjusting specified time period.

4a. If the report generation fails due to technical issues, the system displays an error message and advises the manager to try again later.

Inputs:

- Manager's login credentials.
- Click reports tab to navigate to reports section.
- Selection of the report type.
- Time period/frame for the report.

Outputs:

- Generated report based on selected metrics and time period.
- Download link or display of the report for the manager.

Postconditions: Manager successfully generates a report on key metrics for the specified time period.

b) View Reports:

Brief Description: This use case describes the process of Managers viewing reports to assess workshop efficiency, resource utilization, and customer satisfaction.

Actor: Manager

Preconditions: Manager must be logged into the system.

Basic Flow:

1. Manager navigates to the "Reports" section.
2. Manager selects the type of report to view (e.g., task completion rates, service turnaround times, customer feedback scores/ratings).
3. The system displays the selected report with relevant performance analytics and metrics.
4. The manager reviews the report to assess workshop efficiency, resource utilization, and customer satisfaction.
5. The manager can engage with the report to delve down into information as needed to make informed decisions.

Alternate Flows:

2a. If there are no available reports or the selected report type is not found, the system displays a message indicating that there are no reports to view.

Inputs:

- Manager's login credentials.
- Click reports tab to navigate to reports section.
- Selection of the report type to view.

Outputs:

- Display of the selected report with relevant performance analytics and metrics.
- Manager's engagement with the report to make informed decisions.

Postconditions: Manager successfully views performance analytics and reports to assess workshop efficiency, resource utilization, and customer satisfaction.

2.5 Appointment Management

a) Scheduling Appointments:

Brief Description: This use case describes the process of Customers scheduling appointments by specifying preferred dates, times, and service requirements. They can cancel appointments if necessary.

Actor: Customer

Preconditions:

- Customer is registered and logged in.

Basic Flow:

1. Customer navigates to the appointment scheduling section.
2. Customer selects preferred date, time, and service requirements depending on the availability.
3. Customer submits the appointment.
4. The system checks for appointment conflicts.
5. If no conflicts, the system approves the appointment and notifies the customer.

Alternate Flows:

- 6a. If there are conflicts, the customer must choose another date,
- 6b. Customer selects an alternative date.
- 6c. Customer can reschedule appointments or cancel appointments.

Inputs:

- Customer's preferred date and time.
- Service requirements.

Outputs:

- Scheduled appointment details.
- Confirmation notification to the customer.

Postconditions:

- Appointment is successfully scheduled.
- Customer receives confirmation of the appointment.

b) Reschedule Appointment

Brief Description: This use case describes the process of Customers having the ability reschedule appointments if needed, providing flexibility in managing their schedules.

Actor: Customer

Preconditions:

- Customer is registered and logged into the system.
- Customer has at least one scheduled appointment.

Basic Flow:

1. Customer navigates to the "Appointments" section.
2. Customer selects the appointment they wish to reschedule.
3. Customer chooses the option to reschedule the appointment.
4. Customer specifies the new preferred date and time for the appointment.
5. System validates the new appointment time for availability and conflicts.
6. If the new time is available and doesn't conflict, the appointment is successfully rescheduled.
7. If the new time is unavailable or conflicts, the system suggests alternative times or dates.
8. Customer confirms the rescheduled appointment.

Alternate Flow:

5a. If the new time is unavailable or conflicts, the system suggests alternative times or dates.

5b. Customer selects an alternative time or date.

5c. Go back to step 5.

Inputs:

- Customer's login credentials.
- Selected appointment to reschedule.
- New preferred date and time for the appointment.

Outputs:

- Confirmation of the rescheduled appointment.
- Notification of any conflicts or unavailability with suggested alternatives.

Postconditions:

- Appointment is successfully rescheduled.
- Customer receives confirmation of the rescheduled appointment.

c) Cancel Appointment

Brief Description: This use case describes the process of Customers having the ability to cancel appointments if needed, allowing them to manage their schedules effectively.

Actor: Customer

Preconditions:

- Customer is registered and logged into the system.
- Customer has at least one scheduled appointment.

Basic Flow:

1. Customer navigates to the "Appointments" section.
2. Customer selects the appointment they wish to cancel.
3. Customer chooses the option to cancel the appointment.
4. Customer confirms the cancellation.

Alternate Flow:

N/A

Inputs:

- Customer's login credentials.
- Selected appointment to cancel.
- Optional: Reason for cancellation.

Outputs:

- Confirmation of the cancelled appointment.
- Optional: Confirmation of the reason for cancellation.

Postconditions:

- Appointment is successfully cancelled.
- Customer receives confirmation of the cancelled appointment.
- Optional: Reason for cancellation is recorded.

d) View Appointments:

Brief Description: This use case describes the process of Managers viewing all scheduled appointments to effectively manage the workshop's workflow and resource allocation.

Actor: Manager

Preconditions:

- Manager is logged into the system.
- Appointments are scheduled.

Basic Flow:

1. Manager navigates to the "Appointments" section.
2. System displays a list of all scheduled appointments.
3. Manager reviews the list of appointments, including dates, times, customer names, and service requirements.

Alternate Flow:

N/A

Inputs:

- Manager's login credentials.

Outputs:

- Display of all scheduled appointments with details.

Postconditions:

- Manager successfully views all scheduled appointments.

e) Updating Appointments:

Brief Description: This use case describes the process Managers updating existing appointments in case of changes or conflicts.

Actor: Manager

Preconditions:

- Manager is logged into the system.
- Appointment exists.

Basic Flow:

1. Manager navigates to the "Appointments" section.
2. Manager selects the appointment to be updated.
3. Manager modifies the appointment details (date, time, service requirements).
4. System validates the changes and updates the appointment accordingly.

Alternate Flow:

N/A

Inputs:

- Manager's login credentials.
- Selected appointment.
- Updated appointment details.

Outputs:

- Confirmation of the updated appointment.

Postconditions:

- Appointment details are successfully updated.

d) Create Available Slots:

Brief Description: This use case describes the process of Managers creating available slots for scheduling appointments based on workshop capacity and staff availability.

Actor: Manager

Preconditions:

- Manager is logged into the system.
- Workshop capacity and staff availability are known.

Basic Flow:

1. Manager navigates to the "Slots Management" section.
2. Manager specifies the date, time range, and available services for the slots.
3. System creates available slots based on the specified parameters.

Alternate Flow:

N/A

Inputs:

- Manager's login credentials.
- Date and time range for available slots.
- Available services during the slots.

Outputs:

- Confirmation of the created available slots.

Postconditions:

- Available slots are successfully created for scheduling appointments.

2.6 Skills Sharing

a) Add to Skills Sharing

Actor: Manager

Brief Description: This use case describes the process of manager adding skills to the skills sharing database for every vehicle being serviced. This includes selecting an existing vehicle from the database or creating a new entry for a new vehicle. The manager can specify which part of the vehicle was serviced and provide details on how it was serviced.

Preconditions:

- The manager is logged into the system.

Basic Flow:

1. The manager selects the option to add skills to the skills sharing database.
2. If adding skills for an existing vehicle, the manager searches for the vehicle by name.
3. If the vehicle is found, the manager selects the vehicle model.
4. The manager specifies the part of the vehicle that was serviced.
5. The manager provides details on how the servicing was performed.
6. The manager saves the information to the skills sharing database.

Alternate Flow:

3a. If the vehicle being serviced is not found in the database, the manager creates a new entry for the vehicle by providing necessary details such as make, model, and year.

Inputs:

- Manager's login credentials.
- Option to add skills to the skills sharing database.
- Vehicle name or details if adding skills for an existing vehicle.
- Part of the vehicle that was serviced.
- Details on how the servicing was performed.

Outputs:

- Saved information added to the skills sharing database.
- Confirmation message indicating successful addition of skills.

Postcondition:

- The manager successfully adds skills to the skills sharing database for the specified vehicle, including details on the serviced part and how the servicing was performed.
- If adding skills for an existing vehicle, the information is linked to the respective vehicle entry in the database.
- If creating a new entry for a vehicle, the necessary details such as make, model, and year are captured and stored in the database.

b) View Skills Sharing

Actors: Mechanics, Managers

Brief Description: This use case describes the process Mechanics and Managers viewing the skills sharing information for any vehicle stored in the database. They can search for a vehicle by name and select the model to view the skills associated with that vehicle.

Preconditions:

- The actor is logged into the system.
- The actor has appropriate permissions to view skills sharing information.

Basic Flow:

1. The actor selects the option to view skills sharing information.
2. The actor enters the name of the vehicle they wish to view.
3. The system displays a list of vehicles matching the search criteria.
4. The actor selects the desired vehicle model from the list.
5. The system retrieves and displays the skills sharing information associated with the selected vehicle.

Inputs:

- Actor's login credentials.
- Option to view skills sharing information.
- Name of the vehicle the actor wishes to view.

Outputs:

- List of vehicles matching the search criteria.
- Skills sharing information associated with the selected vehicle model.

Postcondition:

- The actor successfully views the skills sharing information for the selected vehicle.
- If applicable, the actor can access and utilize the skills sharing information to aid in servicing or managing the vehicle.

3. Activity Diagrams

3.1 User Management

a) User login

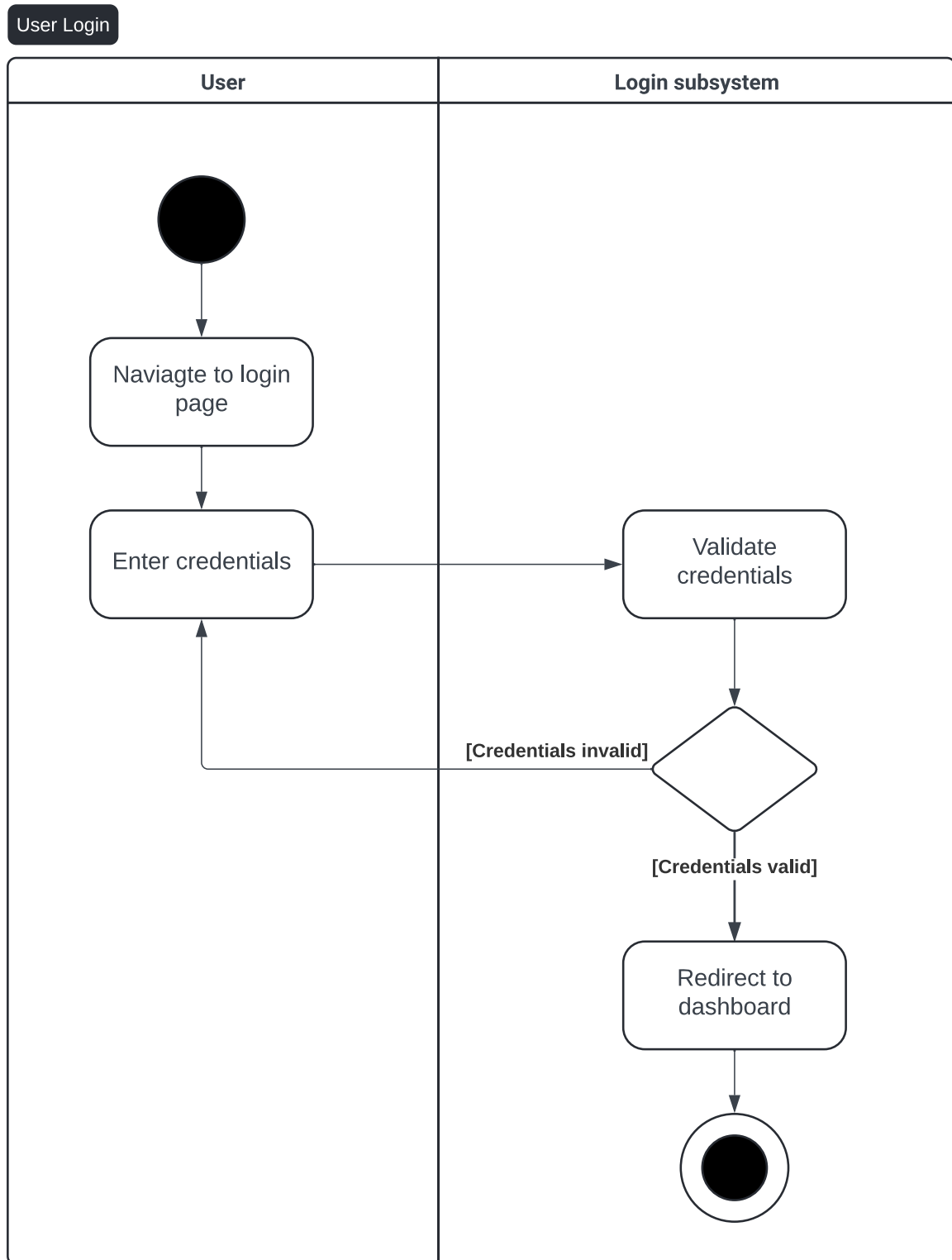


Figure 7: User Login Activity Diagram

b) User Registration

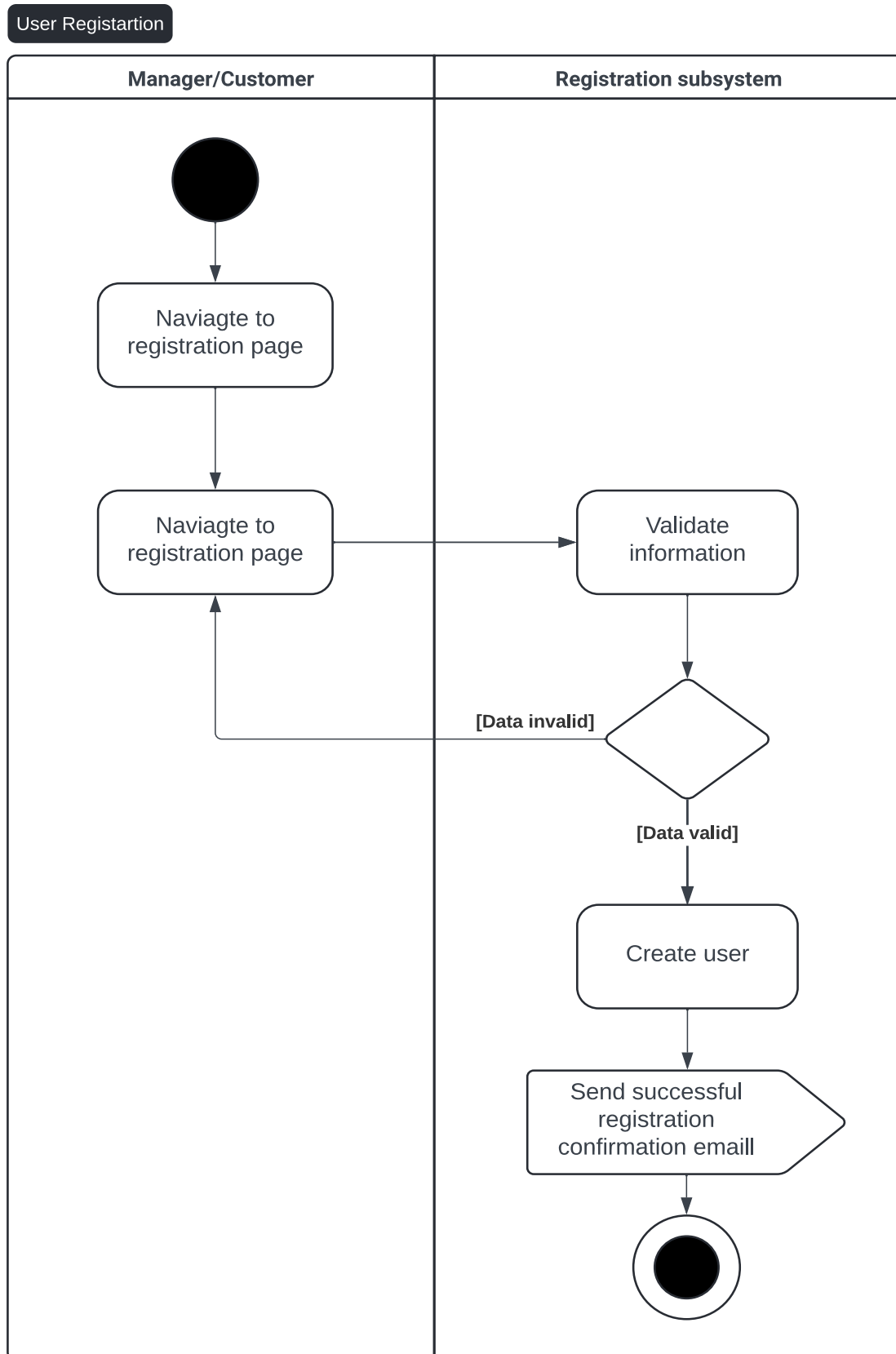


Figure 8: User Registration Activity Diagram

c) Profile Management

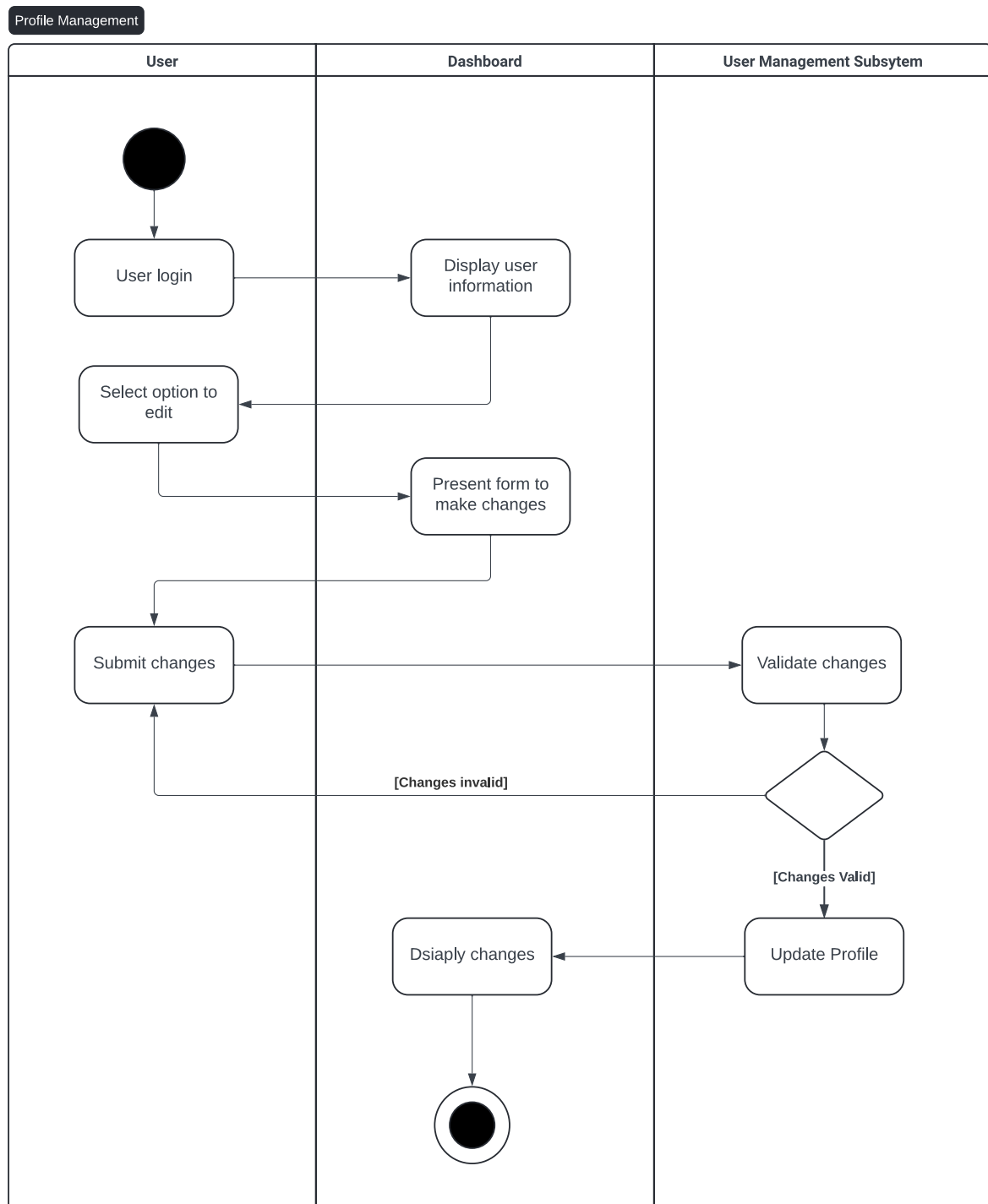


Figure 9: Profile Management Activity Diagram

3.2 Service Request Management

a) Service Request

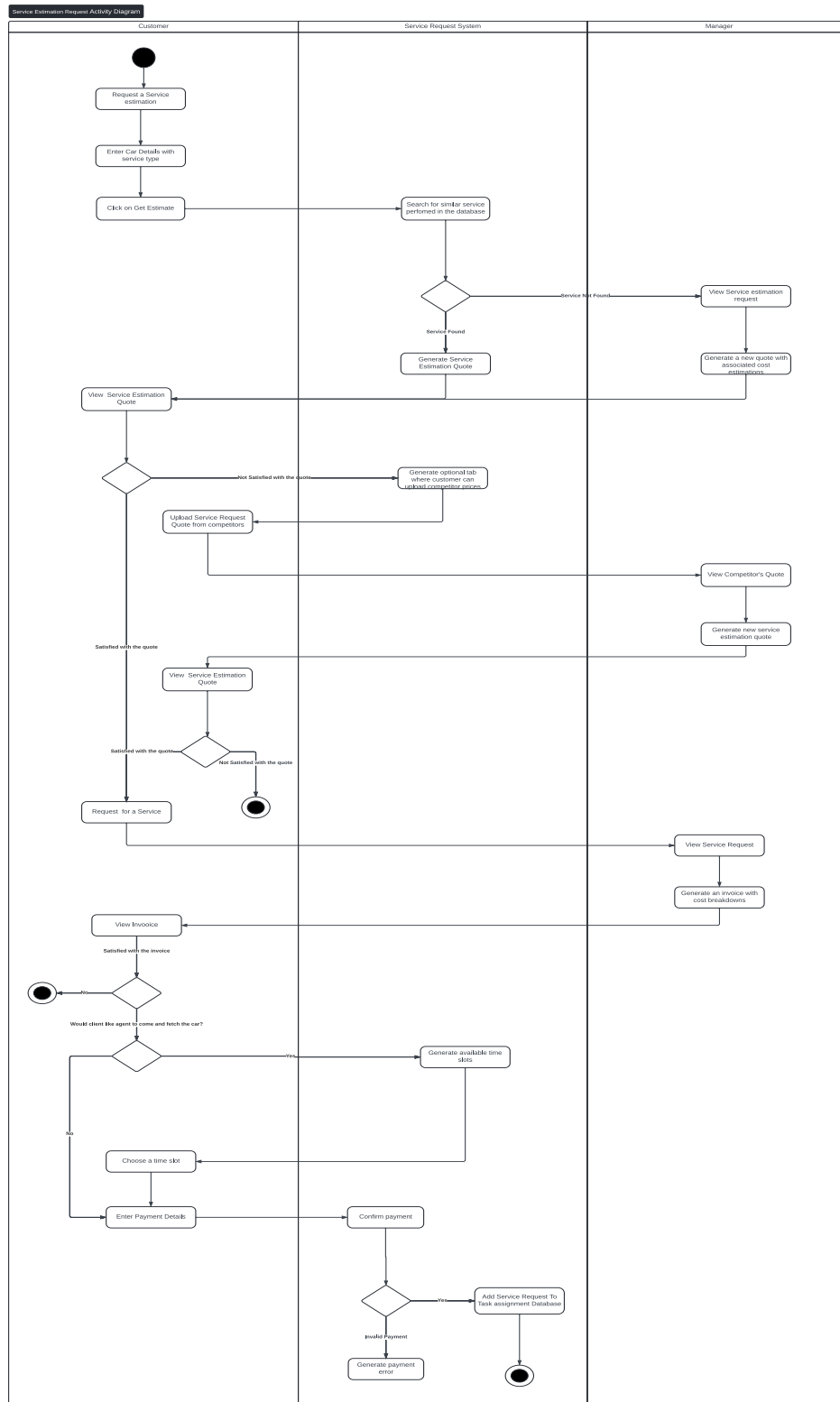


Figure 10: Service Request Activity Diagram

b) Service Request History

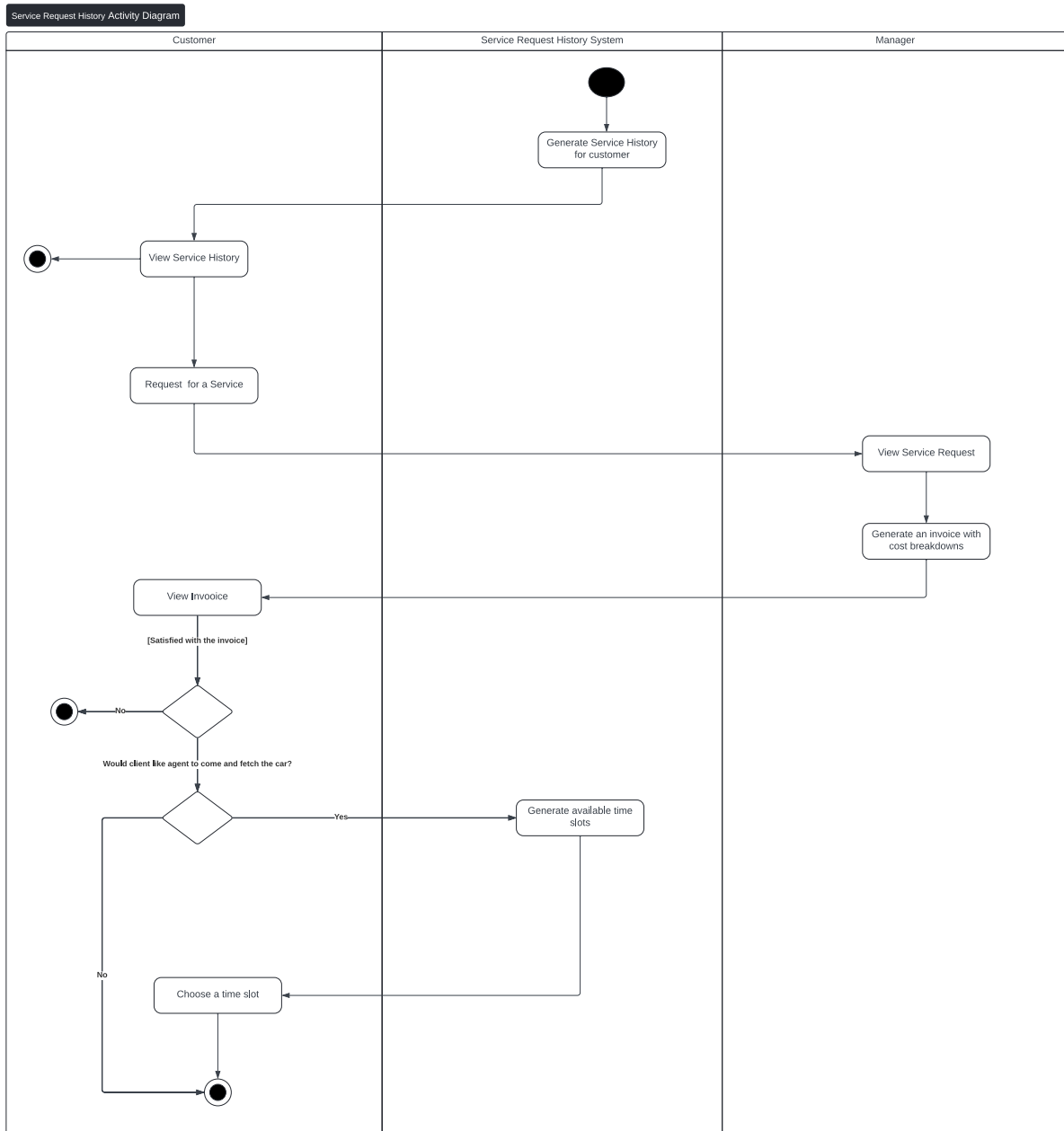


Figure 11: Service History Activity Diagram

3.3 Task management

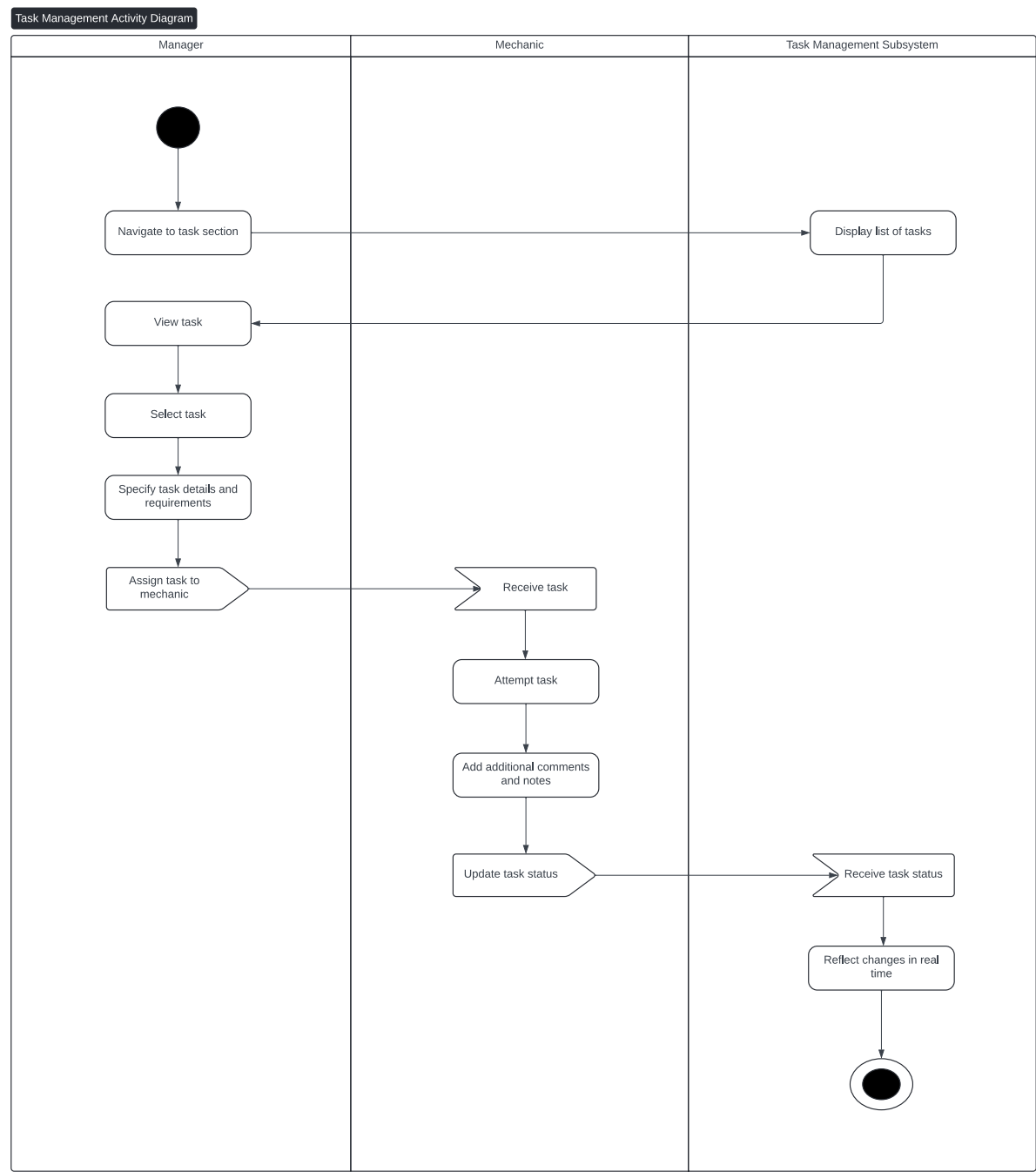


Figure 12: Task Management Activity Diagram

3.4 Performance Monitoring

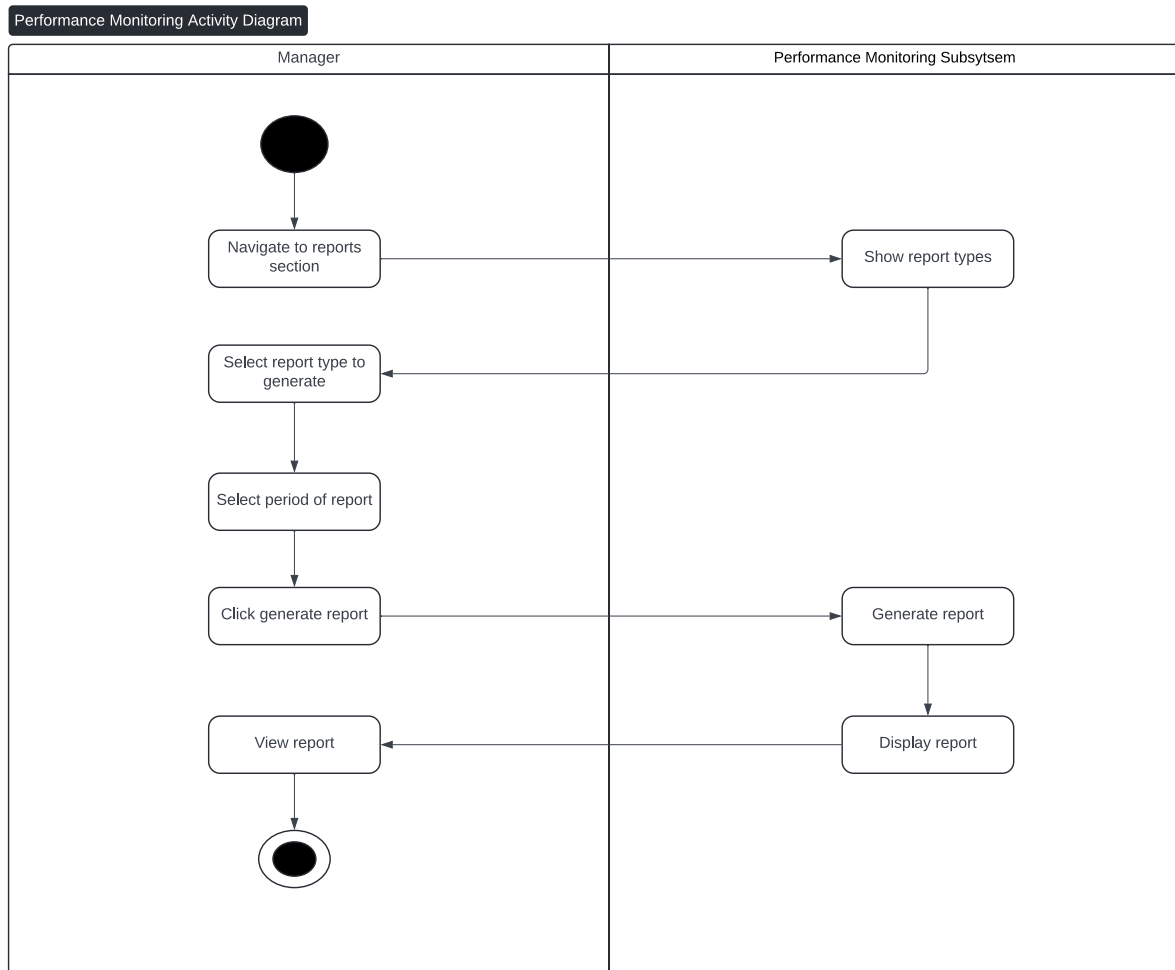


Figure 13: Performance Monitoring Activity Diagram

3.5 Appointment Management

a) Schedule Appointment

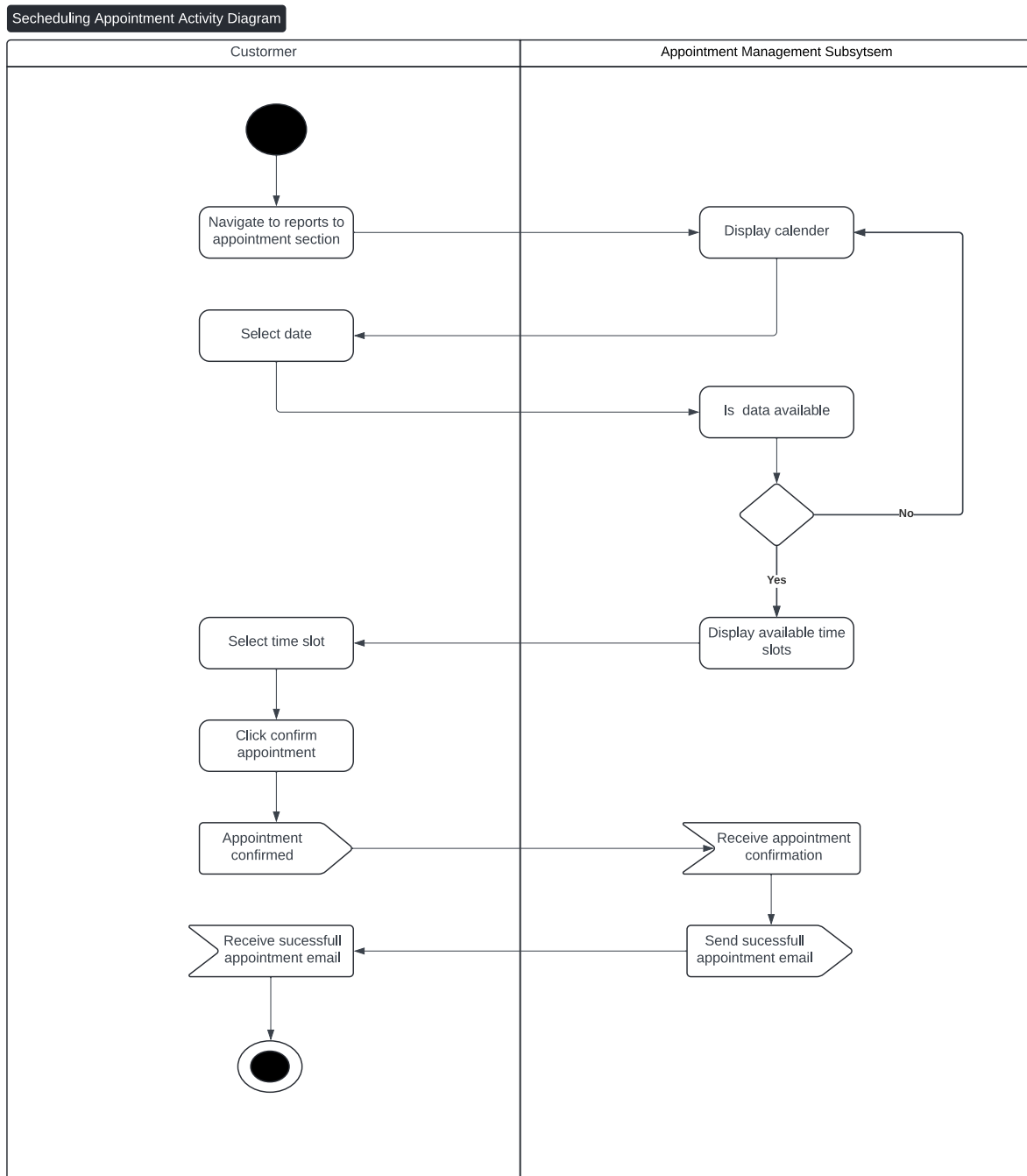


Figure 14: Schedule Appointment Activity Diagram

b) Reschedule Appointment

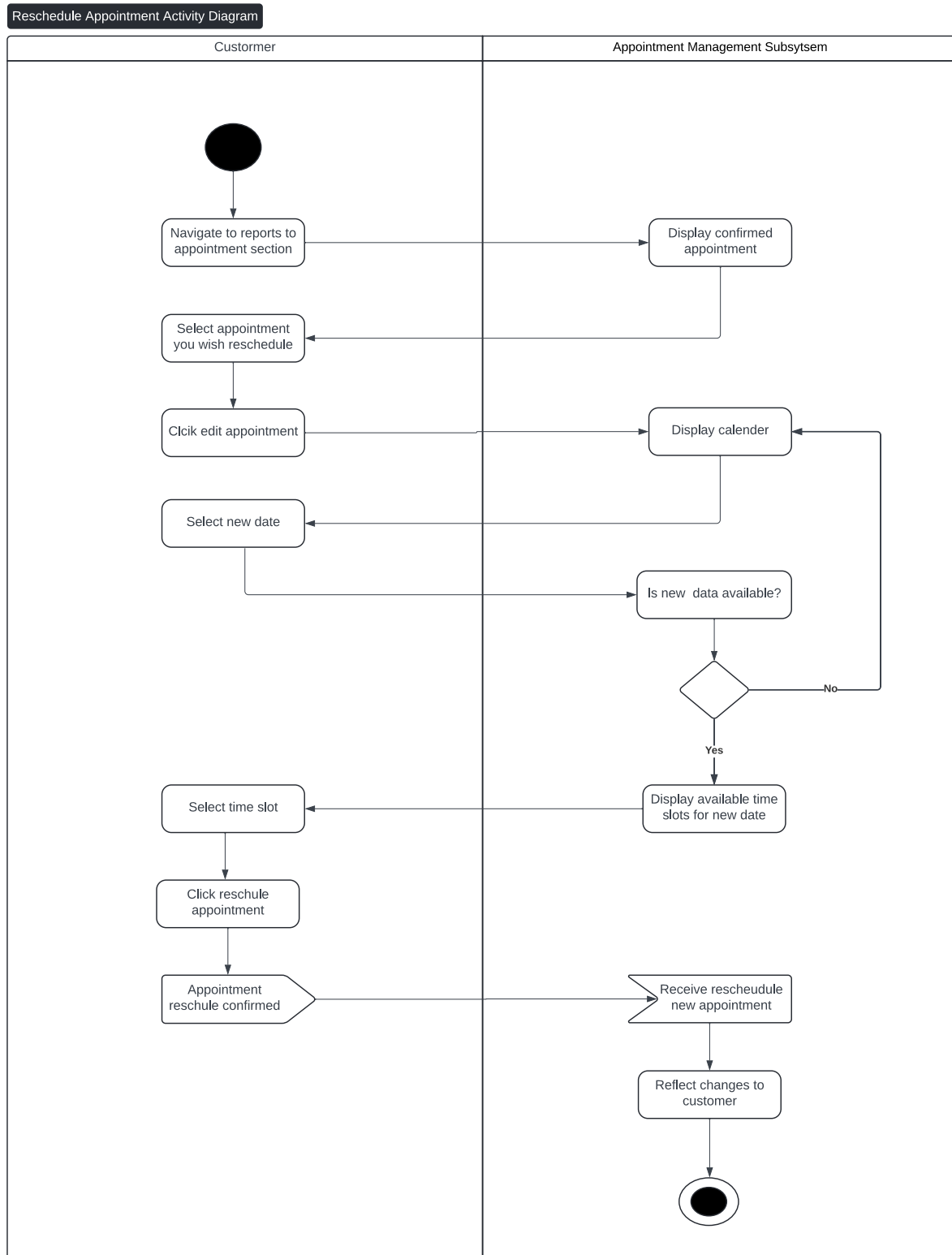


Figure 15: Reschedule Appointment Activity Diagram

c) Cancel Appointment

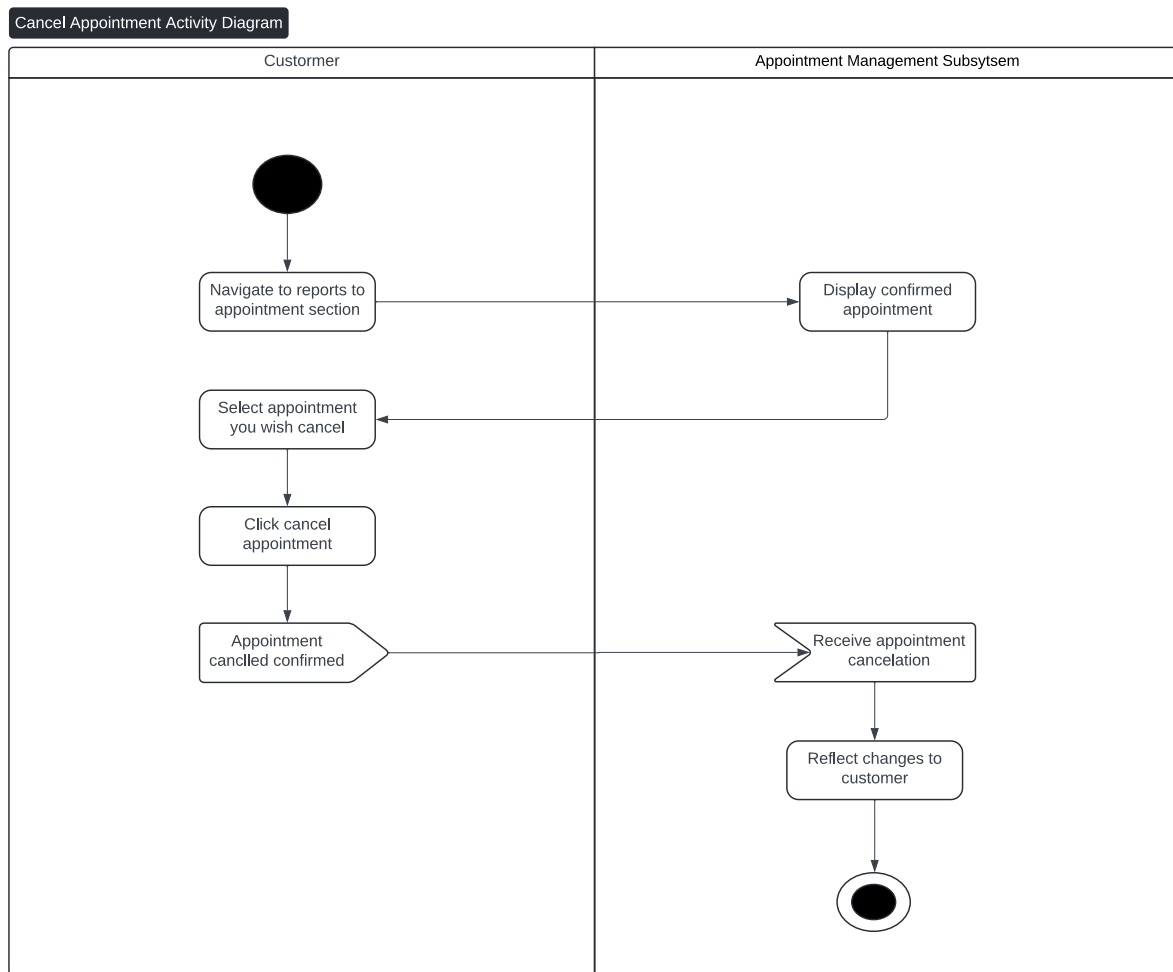


Figure 16: Cancel Appointment Activity Diagram

d) View Appointments:

View Activity Diagram

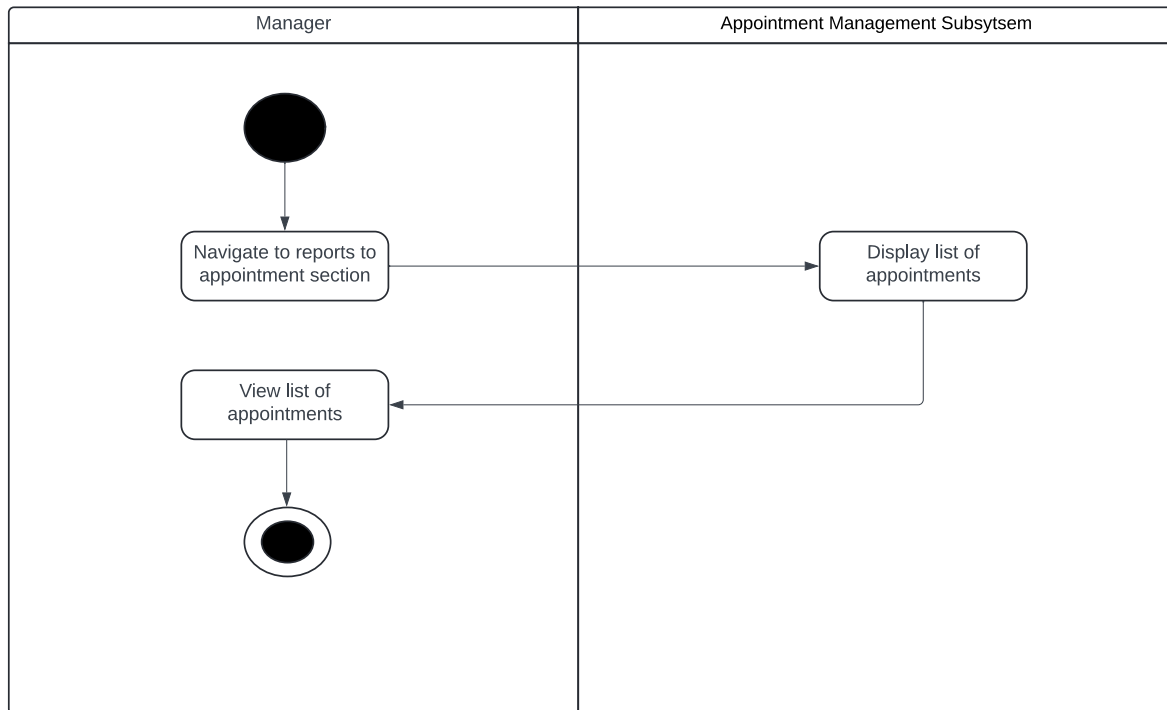


Figure 17: View Appointment Activity Diagram

e) Update Appointment

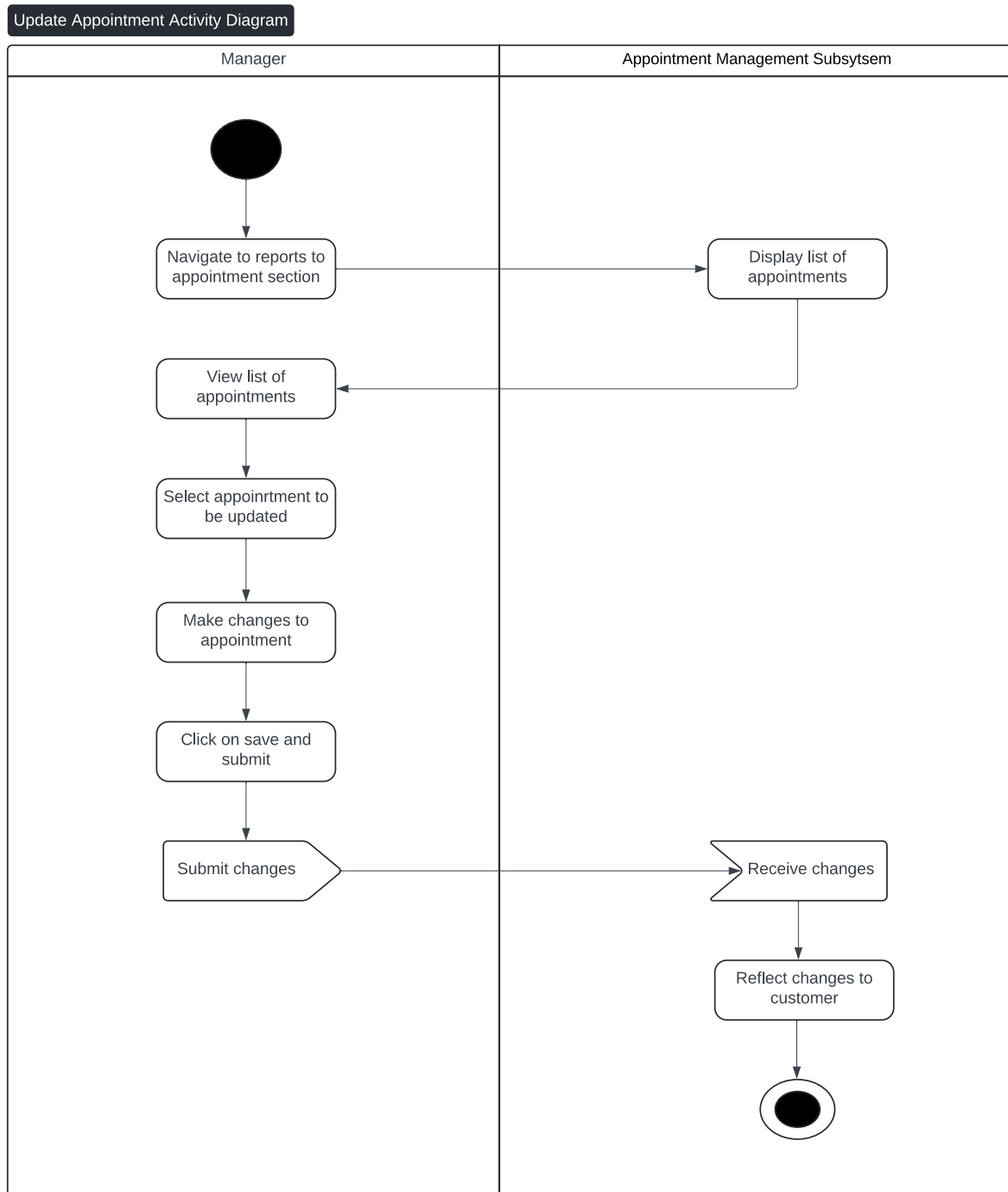


Figure 18: Update Appointment Activity Diagram

3.6 Skills Sharing

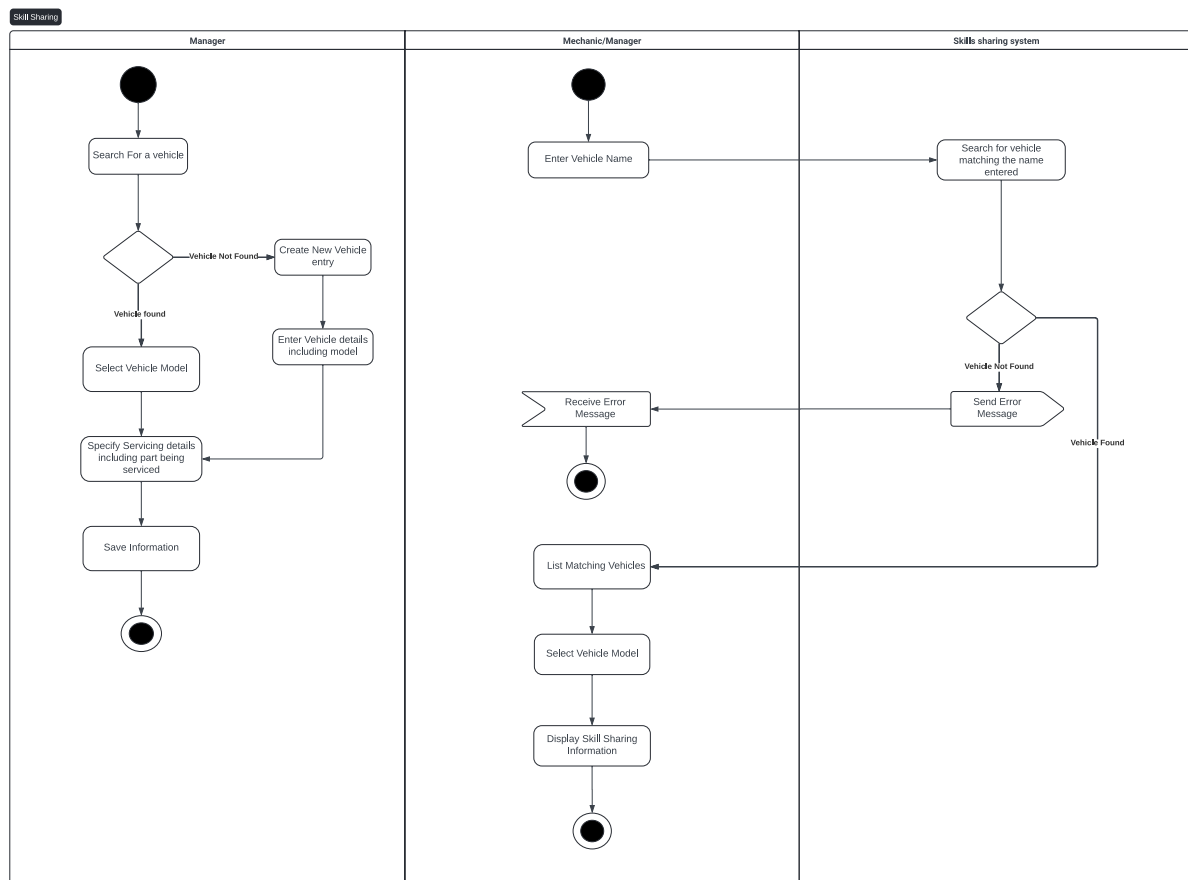


Figure 19: Skills Sharing Activity Diagram

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